

UMESH KHANAL

Full Stack Developer | Applied AI & Machine Learning

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SUMMARY

Full Stack Developer (PERN stack) with hands-on experience building and deploying production web applications, and a parallel track in applied machine learning — computer vision, NLP, and LLM-based tools. Information Science & Engineering graduate (CGPA 9.60/10, University 9th Rank Holder). Comfortable owning a system end-to-end: architecture, UI, backend, deployment, and infrastructure.

SKILLS

Languages: Python, JavaScript, TypeScript, Java, C, HTML, CSS

Frontend: React.js, React Native, Tailwind CSS

Backend: Node.js, Express.js, Django, Flask

Databases: PostgreSQL, MySQL, SQLite

AI / Machine Learning: TensorFlow, PyTorch, Scikit-learn, NLP, Computer Vision (YOLOv11), CNN/LSTM/GRU, LangChain, Google Gemini API

Cloud & Infrastructure: IBM Cloud, IIS, DNS management, SSL/TLS, Windows Server, Vercel

Tools: Git, GitHub, Jupyter Notebook, Google Colab

EXPERIENCE

Full Stack Developer — Rumooz Smart Solutions | Abu Dhabi, UAE | Sept 2025 – Present

- Develop and deploy full-stack web applications using the PERN stack (PostgreSQL, Express.js, React.js, Node.js) with secure RESTful APIs, including JWT and email OTP authentication
- Design responsive, production-grade UIs with Tailwind CSS; implement lazy loading and performance optimization
- Integrate Google Gemini AI agents and chatbots into client websites to automate customer support and improve engagement
- Own end-to-end deployment infrastructure: IIS configuration, DNS management, Windows Server administration, and SSL/TLS setup (see Projects for live production sites)

Machine Learning & AI Intern — Samsung PRISM (R&D Institute, Bangalore) | India | Jan 2024 – Jan 2025

- Worked on prompt engineering and Generative AI projects, contributing to a GenAI – Prompt Generation initiative
- Developed and fine-tuned Large Language Models for practical AI applications, gaining hands-on experience in NLP

PROJECTS

Rumooz Smart Solutions — Production Web Platform rumooz.ae

- Built and deployed using the PERN stack with secure RESTful APIs (JWT + email OTP authentication), a responsive Tailwind CSS UI with lazy loading, and full IIS/SSL deployment with technical SEO

The CUE Group, QHome, QForce, & QEV — Business Websites thecue.ae, ghome.ae, qforce.ae, gev.ae

- Developed and deployed responsive React + Tailwind CSS websites with performance optimization and technical SEO; managed IIS hosting, SSL/TLS configuration, and Google Search Console domain verification for each

Real-Time Violence Detection System (YOLOv11 + GRU)

- Built a real-time surveillance system combining YOLOv11 spatial detection with GRU-based temporal modeling and an attention mechanism; achieved 92% accuracy with low-latency automated alerting

QueryPilot: AI-Powered SQL Query Generator

- Built a natural-language-to-SQL tool using TF-IDF and text-to-SQL modeling, with SQL-injection safeguards and a Flask-based interface

EDUCATION

B.E., Information Science and Engineering / 2021 – 2025

Visvesvaraya Technological University (Cambridge Institute of Technology), Bengaluru, India

— CGPA: 9.60/10, University 9th Rank Holder

CERTIFICATIONS

- Coding and Programming in Python — SIC
- Full Stack Development Certification
- Database and SQL — Infosys Springboard

LANGUAGES

- English (Fluent)
- Nepali (Native)
- Hindi (Conversational)